

Module 3 - Limitations

Q1: Having carried out this assignment, please write two paragraphs about the inherent limitations of carrying out analytics over anonymously submitted data items. Did the analytic responses surprise you?

One of the primary limitations of conducting analytics on Grad Cafe data is that the dataset is based entirely on voluntary self-reporting rather than verified admissions records. Because submissions are anonymous, the data may contain missing information, inaccurate values, inconsistent formatting, duplicate entries, or reporting errors that cannot be independently validated. Also, the dataset is unlikely to be representative of all graduate school applicants because individuals with exceptionally strong, weak, or unusual outcomes may be more motivated to submit their results. The analytic results did not surprise me, but they reinforced the importance of understanding data quality and selection bias when interpreting findings. While the SQL analysis revealed useful patterns, such as a 3.75 average GPA and a 165.81 average GRE Quantitative score, these results should be viewed as trends within the Grad Cafe community rather than definitive admissions statistics.

Q2: How does this differ from standards? For example, the average GRE quantitative reasoning score was 157 for 2023-2023 and was nearly 165 for grad school entries submitted (see sample output). Why do you think that is? What might cause this to occur?

The difference from national benchmarks is likely caused by several forms of selection bias. Applicants with stronger academic credentials may be more likely to report their scores, while those with lower scores may choose not to participate. Additionally, the dataset primarily reflects applicants to graduate programs, many of whom are applying to highly competitive STEM and engineering fields that typically attract students with stronger quantitative backgrounds. As a result, the higher average GRE scores observed in the dataset are likely influenced by self-selection, the characteristics of the applicant pool, and the limitations of anonymous self-reported data rather than representing the broader population of GRE test takers.